AYE
TECH HUB

• INDUSTRIAL ROBOTICS • LESSON 01

Introduction to Industrial Robotics

The Machines That Are Reshaping Manufacturing

Lesson 01 of 30 | Industrial Robotics Program

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- What industrial robots are and how they changed manufacturing
- The 6 major robot types — articulated, SCARA, delta, Cartesian and more
- The 6 degrees of freedom (DOF) — the secret of robot flexibility
- Top robot brands: ABB, KUKA, Fanuc, Yaskawa, Universal Robots
- Industrial robotics career pathways and market demand in Africa

What Is Industrial Robotics?

An **industrial robot** is a programmable, multi-axis mechanical arm designed to perform manufacturing and production tasks automatically — welding, painting, assembly, material handling, packaging, and inspection. Industrial robots operate at speeds, precisions, and endurance levels that no human can match, and they do it safely, consistently, and without fatigue around the clock.

The global industrial robotics market exceeded **$20 billion in 2024** and is growing at 12% per year. Over 4 million industrial robots are currently operating worldwide, with China, Japan, South Korea, Germany, and the USA as the largest markets. Africa is at the beginning of its robotics adoption curve — which means right now is the best time to build this expertise.

The 6 Types of Industrial Robot

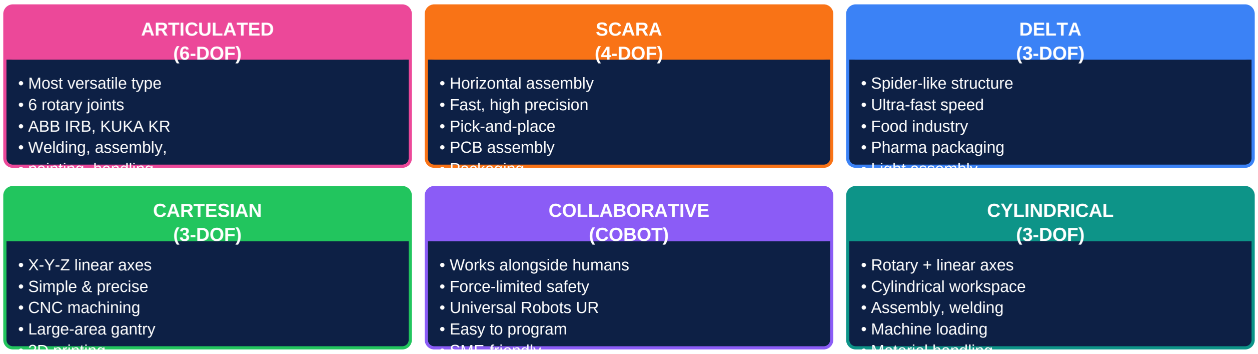

Figure 1 — The Six Main Types of Industrial Robot

Type	Axes	Workspace	Strengths	Typical Application
Articulated (6-DOF)	6 rotary	Spherical	Most flexible, widest reach, handles complex paths	Welding, painting, assembly, machine loading
SCARA	4 (3R+1L)	Cylindrical	Fast, rigid vertically, excellent horizontal precision	PCB assembly, pick-and-place, packaging
Delta (Parallel)	3-4	Dome-shaped	Extremely fast (up to 200 picks/min), light payload	Food sorting, pharma packaging, light assembly
Cartesian (Gantry)	3 linear	Rectangular	Simple, highly accurate, large work volumes	CNC machining, 3D printing, large-area handling
Collaborative (Cobot)	6 rotary	Spherical	Force-sensing safety, works near humans, easy to program	Quality inspection, assembly assist, light handling
Cylindrical	2R+1L	Cylindrical	Simple, reliable, good for rotary+vertical tasks	Machine loading, assembly, spot welding

Choosing the right robot type: The wrong robot type for an application wastes money and causes problems. A Delta robot is perfect for sorting chocolates at 150/min but useless for welding a car chassis. An articulated 6-DOF robot can do almost anything but costs more and takes longer to program than a Cartesian for a simple task.

Degrees of Freedom and Robot Anatomy

A **Degree of Freedom (DOF)** is one independent axis of motion. The human arm has 7 DOF (shoulder 3, elbow 1, wrist 3). A standard industrial robot has 6 DOF — enough to position and orient a tool at any point in its workspace. More DOF = more flexibility but more complexity and cost.

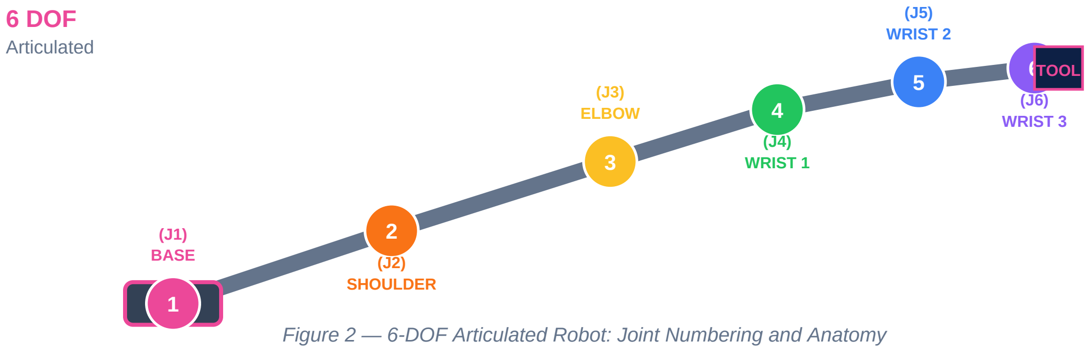

Figure 2 — 6-DOF Articulated Robot: Joint Numbering and Anatomy

Joint	Name	Motion Type	Function in 6-DOF Robot
J1 — Axis 1	Waist / Base	Rotary (yaw)	Swings the entire arm left and right around the base
J2 — Axis 2	Shoulder	Rotary (pitch)	Pivots the upper arm forward and backward
J3 — Axis 3	Elbow	Rotary (pitch)	Bends the forearm up and down relative to the upper arm
J4 — Axis 4	Wrist Roll	Rotary (roll)	Rotates the wrist along the forearm axis
J5 — Axis 5	Wrist Bend	Rotary (pitch)	Tilts the tool head up and down (nod motion)
J6 — Axis 6	Wrist Twist / Flange	Rotary (roll)	Spins the tool (end effector) for final orientation

Robot Specification	Definition	Typical Range
Payload (kg)	Maximum load the robot can handle at full speed and accuracy	3 kg (cobot) to 2300 kg (heavy press tending)
Reach (mm)	Maximum distance from robot base to end of wrist (TCP)	500 mm (desktop) to 4200 mm (large articulated)
Repeatability (mm)	Accuracy of returning to the same point repeatedly (not absolute accuracy)	±0.01 mm (precision assembly) to ±0.5 mm (general handling)
Cycle Time (sec)	Time to complete one full pick-and-place cycle at rated payload	0.3 sec (Delta) to 3+ sec (heavy payload robots)
IP Rating	Ingress protection against dust and liquids. IP67 = dust-tight + water immersion	IP40 (clean room) to IP67+ (food/wash-down environments)
Protection Class	Ex-proof classification for hazardous/explosive environments (ATEX)	Standard, ESD clean-room, ATEX Zone 1/2 (paint booths, oil & gas)

Robot Brands, Programming and Applications

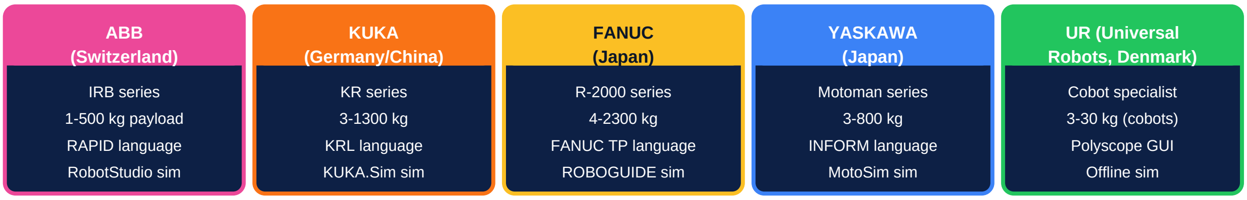

Figure 3 — Top Industrial Robot Brands and Key Specifications

Key Application Areas by Industry

Industry	Robot Type	Application	Robot Count (Global)
Automotive	Articulated 6-DOF	Spot welding, arc welding, painting, assembly, press tending	~1.3M (largest sector)
Electronics / PCB	SCARA, Delta	Component placement, inspection, assembly, solder dispensing	~900K
Food & Beverage	Delta, Articulated	Pick and place, packaging, palletising, sorting by weight/colour	~250K
Metal Fabrication	Articulated	Arc welding, laser cutting, grinding, deburring, machine tending	~400K
Pharmaceuticals	SCARA, Delta	Dispensing, blister packaging, vial handling, sterile assembly	~120K
Logistics & Warehouse	Cartesian, Mobile (AMR)	Order picking, palletising, de-palletising, sorting	~300K (fastest growing)

Industrial Robotics Career Pathways

Role	Core Skills	Salary (Africa, USD/yr)	Demand
Robot Operator / Technician	Basic robot operation, teach pendant, safety procedures	$8,000–$18,000	High — every robot installation needs operators
Robot Programmer	ABB RAPID / KUKA KRL / Fanuc TP programming, simulation	$18,000–$40,000	Very High — skilled programmers are scarce
Robot Systems Integrator	Full cell design: robot + PLC + vision + end effector + safety	$30,000–$60,000	Very High — integration expertise is highly paid
Robotics Application Engineer	Technical sales, feasibility studies, ROI calculation, commissioning	$35,000–$70,000	High — connects customer needs to robot solutions
Robotics Maintenance Engineer	Preventive maintenance, fault diagnosis, servo/motor repair	$20,000–$45,000	High — every installed robot needs ongoing support

What comes next — ROBOT-002: Robot Anatomy and Components — deep dive into the mechanical, electrical and control systems that make up an industrial robot: servo motors, gearboxes, encoders, controllers, teach pendants, and safety circuits.